# SEAM v6.4.9 Validation Release

This release is uniformly synchronized to SEAM v6.4.9. All titles, filenames, scripts, manifests, and document bodies use the same revision.

## Corrected framing

SEAM is a bounded invariant descriptor framework. Hamiltonian notation is a cross-reference layer only. Delta(H->SEAM) measures framework disagreement, not correctness by itself.

## Internal suite result

PASS 7, REVIEW 0; mean composite 90.82%.

## N-body numerical baseline comparison

| Method | Max energy error | Mean energy error | Min pair distance |
|---|---|---|---|
| rk4 | 2.00827e-10 | 7.15477e-11 | 0.18555 |
| verlet | 3.674e-07 | 1.2929e-07 | 0.185542 |
| leapfrog | 3.674e-07 | 1.2929e-07 | 0.185542 |

The integrator outputs are used as external numerical descriptors. This is stronger than self-consistency, but it is still simulation validation, not experimental validation.

## Sensitivity analysis

| Metric | Value |
|---|---|
| Samples | 200 |
| Composite mean | 90.67% |
| Composite std | 0.22% |
| Composite min | 90.13% |
| Composite max | 91.26% |
| Residual mean | 0.00961771 |

## Selected demonstration candidate

The N-body case remains labeled as the selected demonstration candidate, not as a universally proven strongest candidate.

Delta norm: 0.209901; relative delta: 23.35%.

## Limitations

This release still requires real experimental data and domain-specific external baselines for QCD, FCI, SYK, plasma, and turbulence. Coefficients remain provisional unless independently derived or systematically optimized.